

Agentic Coding for Economists

Padova five-hour workshop

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Participant repo: https://github.com/meleantonio/AC4E_Padova

How To Use This Session

Pattern

Short explanation, bounded demo, participant action, verification note.

- Keep [START_HERE.md](#), [SCHEDULE.md](#), and [GUIDE.md](#) open.
- Work from one tool lane: Codex, Claude Code, or Cursor.
- Treat agents as implementation assistants, not research owners.
- Every delegated task ends with evidence: diff, command output, PR, or review log.

Five-Hour Arc



- 00:00-00:45: orientation, safety, agent loop.
- 00:45-01:20: Card-Krueger scaffold.
- 01:30-02:15: skills, subagents, hooks, MCP.
- 02:15-03:00: issues, cloud agents, swarms, loops, goals.
- 03:00-05:00: sprint, verification, adoption.

Participant Repository Path

Read first

- `README.md`
- `START_HERE.md`
- `SCHEDULE.md`
- `GUIDE.md`
- `docs/sources.md`

Use during demos

- `examples/card-krueger/`
- `examples/starter_article/`
- `materials/tool_tracks.md`
- `templates/`
- `slides/workshop/demo_map.md`

What You Produce

- A scoped project brief or design memo.
- Agent instructions and privacy boundaries.
- One usable harness piece: skill, subagent, hook, MCP config, goal, or loop.
- One GitHub issue or cloud-agent task with acceptance criteria.
- One verification artifact: test output, review log, table check, or screenshot.
- A seven-day adoption plan for one real research project.

Human-owned decisions

Research question, identification, data rights, confidentiality, interpretation, and final claims stay with the researcher.

- Do not upload confidential referee material, private data, API keys, or student records.
- Use synthetic or public examples for workshop demos.
- Start read-only; allow edits only after scope is explicit.
- Require commands and outputs before accepting generated code or claims.

First Read-Only Prompt

Demo: orientation without edits

```
Read README.md, START_HERE.md, SCHEDULE.md, GUIDE.md,  
examples/card-krueger/README.md, and  
examples/starter_article/README.md. Do not edit files.
```

Explain:

1. the workshop path;
2. the Card-Krueger running example;
3. which files I should copy or personalize;
4. which files or folders I should not expose to agents.

Question

What happened to fast-food employment in New Jersey relative to eastern Pennsylvania after New Jersey raised its minimum wage in April 1992?

- Treatment group: New Jersey fast-food stores.
- Comparison group: eastern Pennsylvania fast-food stores.
- Timing: before and after the April 1992 policy change.
- Outcome: full-time-equivalent employment.

Public Sources And Teaching Data

- Public source links live in [docs/sources.md](#).
- David Card's page lists the associated public data archive and readme.
- NBER Working Paper 4509 gives the core research context.
- [examples/card-krueger/](#) uses synthetic teaching data so everyone can run the demo immediately.

Caveat

The bundled CSV is not Card and Krueger's raw data and does not support a substantive causal claim.

Data Map Checklist

Required fields

- source and access note;
- unit of observation;
- treatment and comparison groups;
- outcome and units;
- timing and sample restrictions;
- transformations and caveats.

Workshop file

- `examples/card-krueger/`
- `docs/data_source_map.md`
- `templates/data_source_map.md`

Teaching estimand

$$(\bar{Y}_{NJ,after} - \bar{Y}_{NJ,before}) - (\bar{Y}_{PA,after} - \bar{Y}_{PA,before})$$

- The script computes group means and the simple before/after comparison.
- Tests check the input file, group labels, sample counts, command path, and caveats.
- This is a workflow demo, not a replication claim.

Run The Example

Demo: baseline command

```
python3 -m pip install -r examples/card-krueger/requirements.txt
python3 examples/card-krueger/src/did_analysis.py
python3 -m pytest examples/card-krueger/tests
```

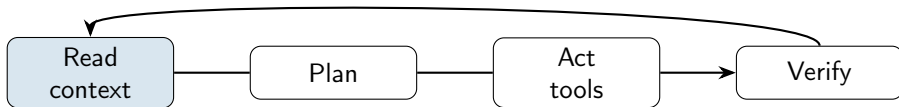
Expected headline:

```
Toy DID estimate: 3.0 FTE workers.
Synthetic teaching data: not Card and Krueger's raw data.
```

From Example To Your Project

- Use [examples/card-krueger/](#) as the shared reference during demos.
- Use [examples/starter_article/](#) as the personalizable harness skeleton.
- Replace the teaching data map with your own data source map.
- Keep the same discipline: scope, evidence, review.

Agent Loop



- Better context improves plans.
- Better acceptance criteria improves edits.
- Better verification improves trust.

Piece	Economist use
Instructions	persistent project map and boundaries
Skills	repeatable procedures for replication, review, writing
Subagents	specialist reviewer contexts
Hooks	deterministic reminders after edits or before risky commands
MCP	structured access to external tools and public data
Goals/loops	longer tasks with stop conditions and evidence

Project Instructions

- Root `AGENTS.md` tells agents how to work in this repo.
- Project copies can use `AGENTS.md`, `CLAUDE.md`, Cursor rules, or all three.
- Include repo map, allowed files, verification commands, and do-not rules.
- Keep instructions short enough that participants can audit them.

Handoff: `templates/AGENTS_economics.md`

Brief, Memo, Spec

- `docs/project_brief.md`: what the project is and is not.
- `docs/research_design_memo.md`: data, design, risks, and claims.
- `spec/intent.md`: optional spec-driven development path for larger projects.
- Agents can draft; humans own identification and interpretation.

Handoff: `templates/project_brief.md`, `templates/research_design_memo.md`

Use when the task is repeatable

A skill packages a procedure that should be invoked consistently.

- replication checker;
- referee checklist;
- paper polisher;
- research SDD plan.

Handoff: examples/starter_article/.cursor/skills/

Subagents And Reviewers

- Use a separate reviewer role for identification, data, literature, or verification.
- Give the reviewer a narrow file set and a blockers-first output format.
- Reviewer agents should not silently fix their own findings.
- In GitHub, request a review after the implementation PR exists.

Handoff: [examples/starter_article/.cursor/agents/](#)

Hooks

- Hooks are deterministic reminders or commands around agent tool use.
- Example: after editing the DiD script, remind the agent to run pytest.
- Keep hook examples safe and explicit; participants trust them before use.
- If a tool version changes hook behavior, check official docs first.

Handoff: [examples/hooks/README.md](#)

MCP And Public Data

- MCP gives an agent structured access to tools such as public-data APIs.
- Use read-only public sources first.
- Put secrets in environment variables, never in the repo.
- For Padova, FRED is the shared optional public-data example.

Handoff: <examples/cursor/.cursor/mcp.json.example>

Goals, Loops, And Stop Conditions

- Use a goal for work that spans several steps.
- Define done criteria before starting.
- Define stop conditions: missing data, failed tests, ambiguous identification, tool limits.
- Record each loop in a review or orchestration log.

Handoff: `templates/long_running_goal_prompt.md`

Choose One Lane

Lane	Tool	Best fit
A1	Codex CLI	terminal-first workflows and GitHub tasks
A2	Codex app	project threads, long goals, visual review
B1	Claude Code CLI	terminal coding and explicit subagents
B2	Claude Code app	app-based code sessions
C	Cursor	IDE-first coding, rules, background agents

Handoff: materials/tool_tracks.md

- Use [AGENTS.md](#) and project-specific instructions.
- CLI and app share the same project files.
- Good for issue-driven implementation, long goals, and PR workflows.
- Verify with commands and attach evidence to PRs.

Handoff: [examples/codex/](#), [examples/codex-app/](#)

- Use `CLAUDE.md` plus shared `AGENTS.md` when collaborating.
- Skills, subagents, and hooks are useful for research review loops.
- Keep project settings in the project, user secrets outside the repo.
- Ask the reviewer role for blockers before fixes.

Handoff: `examples/claude/`, `examples/claude-app/`

- Use rules for persistent project context.
- Use IDE review panes and background agents for branch work.
- Skills and subagents can mirror the same economics harness pattern.
- Verify UI labels and hook support against the installed Cursor version.

Handoff: [examples/cursor/](#)

Version-Sensitive Tool Claims

- Tool surfaces change quickly.
- Do not memorize UI labels from slides.
- Use [docs/sources.md](#) for official docs.
- If a feature is missing, fall back to files and GitHub issues.

GitHub As Control Plane

- Issue: task card with scope and acceptance criteria.
- Branch: isolated implementation.
- PR: reviewable diff plus verification evidence.
- Merge: only after evidence and human review.

Handoff: [orchestration/cloud_agent_issue_template.md](#)

Issue Template For Agents

Title: Add Card-Krueger robustness check

Allowed files:

- examples/card-krueger/src/did_analysis.py
- examples/card-krueger/tests/test_did_analysis.py

Acceptance criteria:

- baseline output remains unchanged;
- robustness result is documented;
- pytest passes;
- synthetic-data caveat stays visible.

Cloud Agents And PR Evidence

- Give cloud agents small, independent issues.
- Require branch name, allowed files, and verification output.
- Do not accept "works" without commands and output.
- Use reviewer agents on the PR, not in the same implementation thread.

Swarm Planning

Stream	Task	Depends on	Use
Data map	source, variables, caveats	none	
Estimation	baseline and tests	data map	
Literature	citations and claim checks	none	
Review	verification and blockers	PRs	

labels: `parallel`, `sequential`, `reviewer`. **Handoff:**
`orchestration/card_krueger_swarm.md`

When Cloud Tools Fail

- Record the failure exactly: permission, missing dependency, usage limit, or build error.
- Use local fallback only for the blocked slice.
- Preserve small commits and PR evidence.
- Return to cloud agents on the next independent issue.

Research Sprint

- Pick one artifact: memo section, test, table, data map, or review log.
- Give the agent one file set and one output.
- Stop after the first useful diff.
- Verify before expanding scope.

Handoff: [notes/orchestration_log_template.md](#)

Verification Stack

- Unit tests for code paths.
- Replication checks for outputs and tables.
- Referee-style review for identification and overclaiming.
- Playwright or screenshots for browser evidence.
- Human sign-off for interpretation.

Handoff: [docs/research_article_harness.md](#)

Verification Prompt

Act as an economics referee. Review only
examples/card-krueger/docs/research_design_memo.md.

Return blockers first:

- identification;
- data source;
- sample definition;
- outcome units;
- overclaiming.

Do not edit files.

Adoption Checklist

- Choose one real project.
- Add an instruction file and privacy boundary.
- Add one data source map.
- Add one verification command or checklist.
- Create one issue with acceptance criteria.
- Run one reviewer pass before trusting output.

Takeaway

Agentic coding is useful for economists when delegation is bounded by context, acceptance criteria, and verification.

- Start small.
- Keep the human in charge of research claims.
- Make every agent task leave evidence.